

AAROHAN 2026

ASCENT OF KNOWLEDGE

Class 12th

Name

Contact Number



BITS Pilani
Pilani Campus

 **APOGEE 2026**

Instructions: -

1. This Question Paper contains 4 comprehensive passages. Each passage is followed by some questions. There will be 25 questions in all.
2. Unless otherwise stated, each question has **ONLY ONE** out of the four choices as the correct answer. Marking Scheme: +4 for correct Answer, -1 for incorrect Answer and 0 for not attempting the question.
3. Some questions are Multiple Answer type questions. +4 will be awarded only when all correct options are marked. -1 for all other cases. 0 is awarded for not attempting the question.
4. The questions will test your skills in Physics, Chemistry, Mathematics and Analytical Reasoning, along with your understanding of the passage.

ALL THE BEST (/ ° - °) /

Space for Rough Work

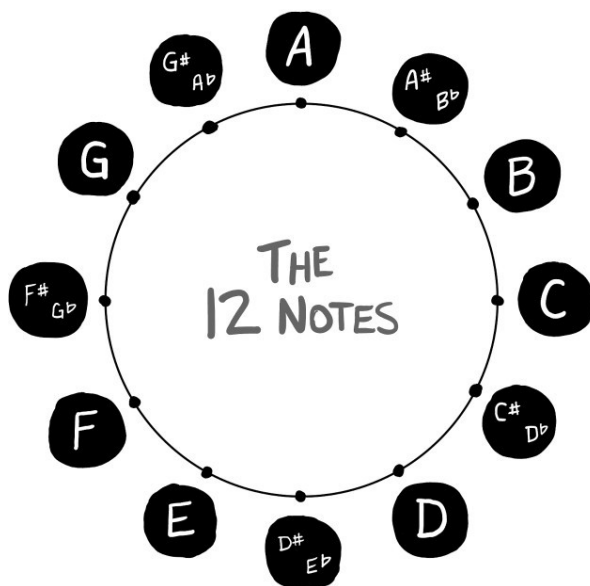
The Sound Of Music

When we hear music, what we actually perceive are vibrations of air. Every vibration has a frequency: how many times per second it repeats. A pure tone (like a tuning fork) produces only one frequency.

To study these frequencies, we can use a graph called a **frequency spectrum**. On this graph, the x-axis is frequency (from low pitches to high pitches), and the y-axis shows **how strong** each frequency is. If you play a flute note at 440 Hz (A4), the spectrum would show a big peak at 440 Hz. This is like taking a “snapshot” of the sound at one instant in time.

In music, certain frequencies are given special names. For example, the note A4 is standardized as 440 Hz, and every octave above it has **exactly double the frequency** (so A5 is 880 Hz). Western music divides the octave into 12 equal parts (**semitones**). This means any semitone step up multiplies the frequency by $2^{1/12}$ (twelfth root of two).

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CLOCKWISE

"UP" or "HIGHER" in PITCH

COUNTER
CLOCKWISE

"DOWN" or "LOWER" in PITCH

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Simple ratios of frequencies, like 3:2, often sound particularly pleasant — this ratio corresponds to a “**perfect fifth**”, a common musical interval.

A **harmonic** of a frequency is any integer multiple of that frequency. For example the second and third harmonics of a note at frequency 100 Hz would be 200 Hz, 300 Hz, respectively. Whenever an instrument plays any note, it also excites the harmonic vibrations of that note along with it. Different instruments do this in different ways, which is why a guitar sounds different from a flute.

If required in any question, the western music octave has the following notes: *A, A#, B, C, C#, D, D#, E, F, F#, G, G#*

Here (#) refers to a “sharp” note. Each of these notes is one semitone higher than the previous one.

[Useful Constants: $\sqrt{2} = 1.414$, $\log_2 1.5 = 0.585$, $2^{(0.67)} = 1.5874$]

Q1. An A4 note is 440 Hz. What would be the frequency of the D# note of the octave corresponding to A5? (Nearest integer) [$\sqrt{2} = 1.414$]

- A. 660 Hz
- B. 622 Hz
- C. 1866 Hz
- D. 1244 Hz

Q2. Which harmonic(s) of A4 (440 Hz) line(s) up exactly with a higher octave of A4? **[Multiple options may be correct]**

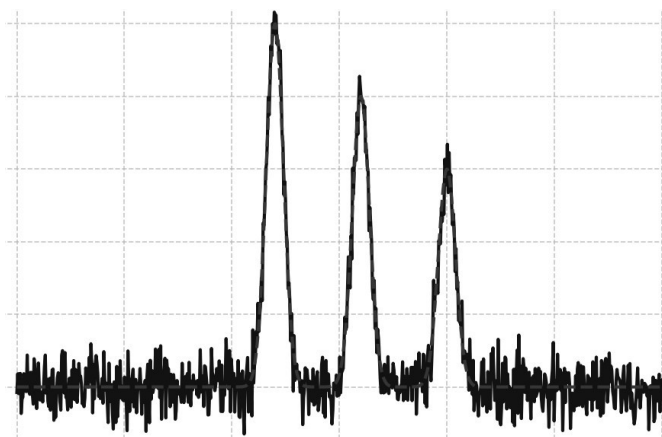
- A. 660 Hz
- B. 880 Hz
- C. 1320 Hz
- D. 1760 Hz

Q3. How many semitones apart would 2 notes be if they are classified as a perfect fifth? Approximate to nearest integer. [Use $\log_2(1.5) = 0.585$].

- A. 7
- B. 5
- C. 3
- D. 6

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Q4. Consider the **augmented C chord**, which consists of the notes C-E-G#. Below is the frequency spectrum of this chord. As expected it has 3 peaks corresponding to these notes. Notice that the distance between consecutive peaks is constant. Identify what kind of scale the x axis has:



- A. Linear because the 3 frequencies are in an arithmetic progression
- B. Logarithmic because the 3 frequencies are in an arithmetic progression
- C. Logarithmic because the 3 frequencies are in a geometric progression
- D. Linear because the 3 frequencies are in a geometric progression

Q5. Which statement is true about semitone steps as frequency rises?

- A. The absolute Hz difference between adjacent semitones is the same anywhere on the keyboard.
- B. The absolute Hz difference between adjacent semitones increases as frequency increases
- C. The absolute Hz difference decreases as frequency increases
- D. The Hz difference doesn't follow a specific trend with respect to frequency

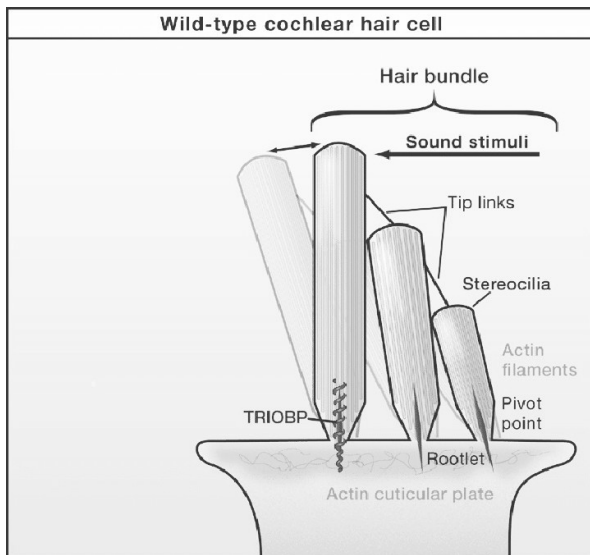
Q6. If the frequencies of a chord's three notes are in a geometric progression with common ratio r , then the semitone step between adjacent notes is constant. For an augmented triad the step is 4 semitones. If the bottom note is $C4 = 261.63$ Hz, what is the top note's frequency (the third note; two steps up)? (choose nearest answer)

- A. 392 Hz
- B. 466 Hz
- C. 415 Hz
- D. 523 Hz

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I Can't H(Ear) You

Inside your inner ear lies a thin membrane that vibrates when sound waves arrive. These vibrations are transferred to **tiny hair-like structures called stereocilia**, which are connected at their tips by delicate **tip link proteins**. When the stereocilia bend from vibration of sounds, these tip links pull open **ion channels** on the hair cell surface, allowing positively charged ions (like potassium or calcium) to rush into the cell. This sudden ion movement changes the cell's electrical state, triggering the conversion of mechanical vibration into an electrochemical signal. This process is called **mechanotransduction**.



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Hair cells maintain a specific electrochemical gradient so that ions can quickly flow in when the channels open. This gradient is sustained by **ion pumps**, which use ATP (energy) to actively push certain ions out while pulling others in *against the concentration gradient*. The correct **shape and arrangement** of the tip link proteins are critical, even a small structural change can prevent the ion channels from opening.

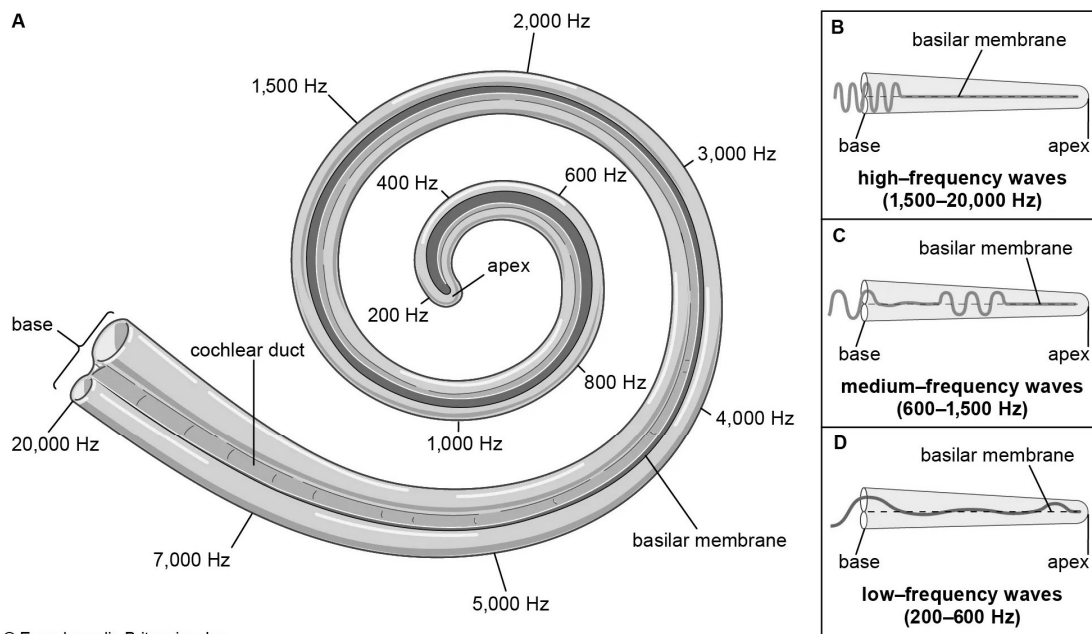
When positive ions rush into the hair cell due to bending of stereocilia, it releases a chemical messenger called **glutamate**. This messenger acts like a signal light — the more sound there is, the more glutamate is released. Nearby nerve cells have **receptors whose ligands bind to glutamate** and use this as the “go” signal to open their ion channels to transmit electrochemical signals to carry sound information to the brain.

Some chemicals can damage these delicate structures of hair cells. For example, **cisplatin**, a chemotherapy drug, is effective at killing cancer cells but can also **harm hair cells**, leading to **ototoxicity** (hearing loss caused by chemical damage). Scientists believe cisplatin can disrupt the proteins or the ion transport processes within these cells, making them less able to convert sound vibrations into electrical signals.

Inside the cochlea lies a thin, flexible structure called the **basilar membrane**. Its main job is to separate and detect different sound frequencies. What makes it special is that its physical properties are not the same everywhere: near the base of the cochlea it is **stiff and narrow**, while toward the far end (the apex) it **becomes wider and much more flexible**. Because of this design, **high-pitched** sounds (like a whistle) cause the stiff **base** to vibrate, whereas **low-pitched** sounds (like a drumbeat) travel deeper and set the flexible **apex** into motion. This gradual change along its length allows the basilar membrane to distinguish between frequencies. Let's assume (hypothetically) that the relation between frequency and distance along the basilar membrane is given by:

$$f = 20,000 (e^{-6.909x})$$

Here, x is the fraction of distance along the basilar membrane *from the base*. So $x = 0$ for the base and 1 for the apex.



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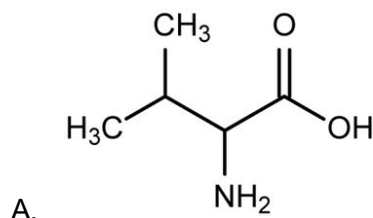
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Q7. Tip link proteins connect adjacent stereocilia and are under constant tension, so they must resist mechanical stress without breaking. Which of the following structural features *would* most likely contribute to their function?

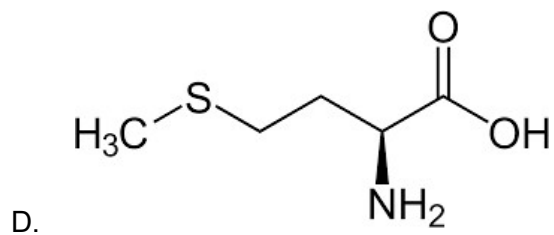
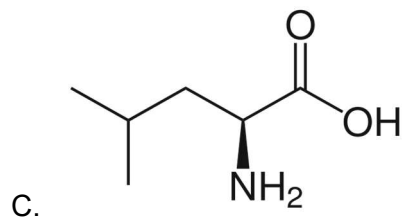
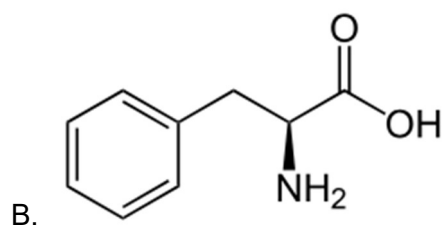
- A. A high proportion of layered sheets, with extensive hydrogen bonding between strands
- B. A loosely coiled helical shape with few intramolecular hydrogen bonds
- C. A high concentration of nonpolar side chains forming a hydrophobic region
- D. Randomly oriented peptide chains with minimal secondary structure

Q8. Let's say that cisplatin, a cancer drug, can cause permanent hearing loss by binding to certain amino acids in proteins of the inner ear's hair cells, disabling their antioxidant defenses. Platinum in cisplatin forms coordinate bonds with atoms having higher electron density around them, but not with the atoms of carboxylic acid or amino group.

Based on this, which of the following amino acids is most likely a binding site for cisplatin:



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Q9. Prolonged exposure to very loud sounds can damage auditory nerve fibers. What is the main cause of this damage?

- A. Loud sounds cause large vibrations of tip link proteins, breaking them down into their amino acid components.
- B. Too much glutamate overstimulates receptors, damaging nerve endings
- C. Hair cells run out of potassium and stop sending signals
- D. The eardrum gets damaged from overstimulation

Q10. A mutation prevents hair cells in the cochlea from releasing glutamate when they are stimulated. Which step in the hearing process would be directly disrupted?

- A. Conversion of sound vibrations into bending of stereocilia
- B. Entry of sound waves into ear canal
- C. Bending of tip-link proteins to open ion channels
- D. Transmission of electrical signal from hair cells to auditory nerve

Q11. Use your knowledge from the first paragraph about the notes in an octave. What would the distance be (in terms of x) between the 2 points on the basilar membrane corresponding to the notes A and D# (from the same octave)?

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- A. $6.909/6$
- B. $\ln(6.909/3)$
- C. $\ln(2)/6.909$
- D. $\ln(2)/13.818$

Q12. Consider a graph which plots the Potassium current (I) on Y-axis through open K^+ channels against membrane potential (V) on X-axis under two different conditions:

- 1) Curve A: Normal hair cell with active ion pumps working to maintain gradients.
- 2) Curve B: Abnormal hair cell with inhibition of ATP-dependent pumps after a long time.

(For an open channel, the I-V relationship is roughly linear)

Consider the following statements about these 2 curves and choose the correct option(s) **[Multiple options may be correct]**:

- A. Curve A intersects the X-axis when membrane potential is equal to equilibrium potential.
- B. After the ATP-dependent pumps are inhibited, the equilibrium potential shifts away from 0 mV.
- C. For membrane potentials negative to equilibrium potential, K^+ current is inwards.
- D. The slopes of both the curves are nearly the same.

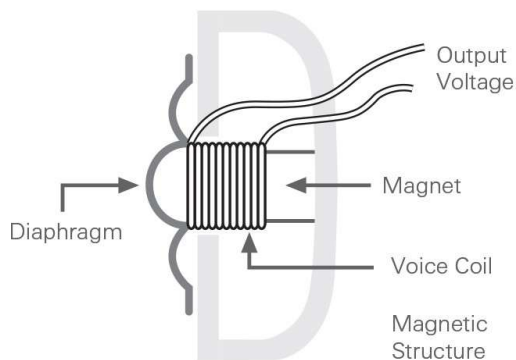
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I CAN HEAR YOU NOW!

When you listen to music or a live performance through an electronic system, the sound goes through a chain of devices. It usually starts with a microphone, which converts pressure variations in the air (sound waves) into small electrical signals. Microphones are of many types, the two most common being **condenser microphones** and **dynamic microphones**.

In the former, sound waves move a **small metallic diaphragm** which acts as one plate of a capacitor. The other plate is fixed. Varying distance between the plates varies the capacitance, and these changes are used to measure the audio signal. In a dynamic microphone, **pressure waves move a diaphragm** which moves a coil in a magnetic field. Through **electromagnetic induction** a varying voltage is produced which is the required audio signal in electrical form. The diaphragm + coil combination is heavier than the diaphragm in condenser mics.

Devices like microphones and speakers that convert one form of energy to another are known as **transducers**.



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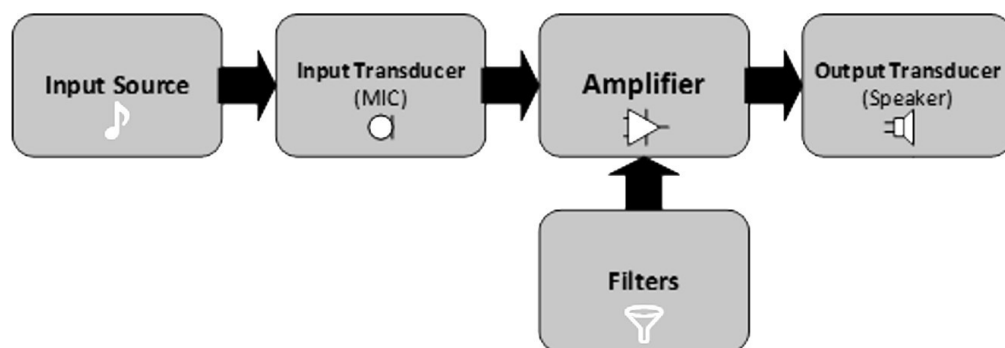
Audio signals received by a mic are very weak, so they are passed into an amplifier. An amplifier increases the voltage or power of a signal, but it doesn't behave the same way at all input levels: in the **linear region**, the output is a clean, scaled-up version of the input, but if the input is too strong the amplifier enters the **saturation region**, where it can no longer increase proportionally, and distortion is introduced. This is due to limitations on the characteristics of transistors involved in the amplifier.

Up to now we looked at spectra with a few sharp peaks corresponding to musical notes, but in real-world audio signals (like speech or music), the spectrum is not just discrete; it can spread **continuously across a whole range of frequencies**. After amplification, the sound is often **shaped using filters**. For example, bass, mid, and treble knobs on a music system are essentially filters that emphasize or suppress different frequency bands.

Beyond filtering, effects such as echo or reverb can be added by **combining the signal with delayed versions of itself**, creating a sense of depth or space. Finally, the signal reaches a speaker, which converts the electrical waveform back into pressure variations in air, completing the chain from sound to electricity and back to sound.

A loudspeaker system is typically composed of multiple specialized **drivers** designed to handle different parts of the frequency spectrum. One is the **tweeter** which plays high frequencies, and the other is the **subwoofer** which is optimized for lower frequencies. Only certain frequency ranges must be sent to each driver by using appropriate filters.

Let's go step by step through the different components in a sound system chain and solve some of the challenges that might show up at each stage.



[Useful constants: $\log_{10} 2 = 0.301$, $\sqrt{10} = 3.162$]

Q13. Choose the most appropriate statements comparing dynamic and condenser microphones.

[Multiple options may be correct]

- A. For a voltage to be measured across the capacitor in a condenser mic, it must store some charge and therefore requires a power supply
- B. Dynamic microphones are more sensitive to external sound
- C. The lightweight condenser diaphragm makes it more responsive to higher frequencies
- D. A dynamic microphone is less responsive to higher frequencies because of inertia of its heavier diaphragm

Q14. Next, let's talk about amplification. An audio amplifier drives a loudspeaker with an input power of 1W. The amplifier has a voltage gain of 4 (i.e., it multiplies the input voltage by 4) and operates at constant load impedance.

1. What is the corresponding power gain (ratio P_{out}/P_{in})?
 2. What is the gain in decibels?
 3. If instead, the amplifier output power were doubled, by how many dB would the gain increase?
-
- A. Power gain = 4, 6 dB, +3 dB increase
 - B. Power gain = 16, 9 dB, +3 dB increase
 - C. Power gain = 4, 6 dB, +6 dB increase
 - D. Power gain = 16, 12 dB, +3 dB increase

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Q15. A guitarist plays two strings. The first vibrates at 220 Hz, and the second vibrates at 330 Hz. Assume that only the second harmonics are excited along with the fundamental frequency, and that they are excited at the same amplitude as the fundamental. Ignore all other higher harmonics.

Now these vibrations are picked up by a receiving mic and the electrical signal is passed through a low-pass filter whose cutoff frequency is 400 Hz. Assume that the frequencies below the cutoff are unaltered. Between 400-600 Hz, the amplitude is reduced by 10 dB, and beyond 600 Hz it is reduced by 12 dB.

The final output is displayed on a frequency spectrum showing frequency on x-axis and amplitude of that frequency on y-axis on a log scale. Which statement best describes the output spectrum?

- A. Both fundamentals are present with equal amplitudes, while both second harmonics are 10 dB lower than the fundamentals.
- B. Both fundamentals are unchanged. The first string's harmonic is reduced by 12 dB while the second string's harmonic is reduced by 10 dB.
- C. The second string's fundamental and harmonic are both present at equal amplitude. The first string's fundamental is unaltered but its harmonic is suppressed by 10 dB.
- D. Both fundamentals are unaltered. The first string's harmonic is attenuated by 10 dB while the second string's harmonic is attenuated by 12 dB.

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Q16. Different audio effects can be achieved in electronic systems in different ways. For example, a reverb effect is achieved when delayed and attenuated copies of the signal are added together.

A digital audio processor adds two delayed copies of the input signal $x(t)$ to simulate reverberations. The second (more delayed) copy is delayed by τ seconds compared to the original signal. Assume equal delay between the original and the first copy, and the first and the second copy. Each delayed copy has its amplitude reduced by a constant factor α compared to the immediately previous copy.

Write an expression for the output signal $y(t)$ in terms of the original signal and the given variables.

- A. $y(t) = x(t) + \alpha x(t - \tau) + \alpha^2 x(t - 2\tau)$
- B. $y(t) = x(t) + \alpha x(t - \tau) + \alpha^2 x(t - \tau)$
- C. $y(t) = x(t) + \alpha x(t - \tau/2) + \alpha x(t - \tau)$
- D. $y(t) = x(t) + \alpha x(t - \tau/2) + \alpha^2 x(t - \tau)$

Q17. For the audio processor described in the previous question, assume that the original signal $x(t)$ is at an amplitude of 22 dB. Assume that human hearing fails to detect sounds below 2 dB. What is the lowest value of α for which we can hear the copy of the signal which is delayed by τ seconds?

- A. 0.3162
- B. 0.1
- C. 0.25
- D. 3.162

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Q18. Consider two drivers within a speaker, A and B. Both of these can be thought of as driven harmonic oscillators whose natural or resonant frequency determines which region of the spectrum they are best suited for.

A has a higher mass and moderate spring constant k while B has a moderate mass but a high spring constant.

Now, there are two filters. One high pass and one low pass filter. Which filter should be used to preprocess the sound signal before it is sent to each speaker driver?

- A. Low pass filter before driver A because A is the subwoofer. High pass filter before B because B is the tweeter.
- B. Low pass filter before A because A is the tweeter. High pass filter before B because B is the subwoofer.
- C. High pass filter before driver A because A is the subwoofer. Low pass filter before B because B is the tweeter.
- D. High pass filter before driver A because A is the tweeter. Low pass filter before B because B is the subwoofer.

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You Wouldn't Have Heard These

Across history, inventors have created instruments that turn physical principles into sound, merging technology and creativity in surprising ways. The following sections explore some of these remarkable designs.

Theremin:

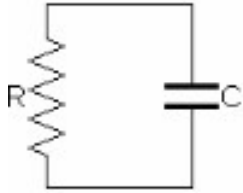
A theremin operates using the principle of **LC oscillators**. One oscillator has a fixed frequency f_1 , while the other, f_2 , varies based on the performer's hand position near a metal antenna. The player's **hand movement changes the distance** d between capacitor plates. This changing capacitance alters the LC oscillator's frequency $\omega_2 = \frac{1}{\sqrt{LC}}$. The difference $|f_1 - f_2|$ determines the frequency of the sound produced by the instrument.

Laser Harp:

A laser harp replaces strings with **laser beams**. **Photodiodes** detect when a performer's hand **blocks a specific beam**. The photodiode current depends on the laser's intensity, which falls sharply when the beam is interrupted. This current is smoothed by an RC circuit. A threshold detector is triggered when the output voltage from the RC circuit **falls below some critical value**, thus playing the corresponding note. In the RC circuit shown below, the output voltage V is taken across the capacitor.

The value of output voltage when the laser beam is uninterrupted is V_0 .

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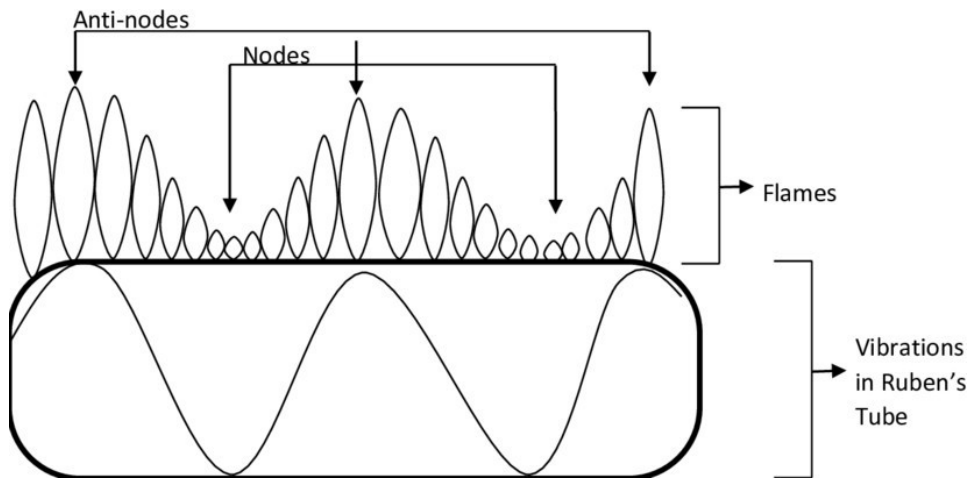
Ruben's Tube:

A Rubens' tube is a long metal pipe **sealed at both ends**, filled with propane that escapes through small holes along its top, where it burns as evenly spaced flames. The gas inside is initially kept at a constant pressure $P_0 = 50 \text{ kPa}$. When a sound wave is introduced at one end, it sets up a **standing pressure wave** inside the tube.

The rate of gas flow through each hole depends on the **local pressure difference**, thus the **flame height** (which depends on this rate) varies along the tube depending on the standing wave.

The total pressure inside the tube varies as $P_0 + P(x, t)$ where the second term is from the pressure variation in the standing wave.

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Hydraulophone:

The hydraulophone creates tones by **directing water through tuned pipes**, where fluid dynamics, Bernoulli's principle, and flow equations determine pitch and intensity. The pressure drop across a hydraulophone key can be modeled as $\Delta P(t) = k_1 x(t) + k_2 v(t) + k_3 a(t)$ where $x(t)$ is the displacement of the key, $v(t)$ is the velocity, $a(t)$ is the acceleration, and k_1, k_2, k_3 are fixed positive constants. The intensity of sound produced is proportional to the **square of this pressure drop**.

Singing Bowl:

The singing bowl, made from a seven-metal alloy, gets its unique sound from the formation of **intermetallic phases** between Cu and Sn. The γ -Brass type phase, with approximate composition

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$Cu_{41}Sn_{11}$ is the composition of material the singing bowl is coated with to create resonance and beating tones.

AudioCube:

The AudioCube is a cube-shaped sound device that uses **proximity** sensors to generate audio. Each of its faces can detect nearby objects, allowing users to **manipulate musical parameters through movement**.

Based on the above paragraphs, answer the following questions:

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Q19. Theremin: Assume that frequency of the fixed LC oscillator is 260 KHz. For the variable oscillator, say $L = 10 \text{ mH}$. *Given that moving your hand further from the antenna reduces the frequency of sound produced*, for what value of d does the Theremin produce a sound at 20 KHz?

[For simplicity, assume the capacitor is a parallel plate capacitor with free space between plates ($\epsilon_0 = 8.854 \times 10^{-12} \text{ F/m}$) and effective area $A = 0.02 \text{ m}^2$.]

[Take $1/150720^2 \approx 5\epsilon_0$, $\pi = 3.14$]

- A. 4 mm
- B. 2.5 mm
- C. 2.5 cm
- D. 2 mm

Q20. Laser Harp: Suppose the threshold detector is triggered when the voltage output drops below $0.3 V_0$. What is the minimum time that a laser beam must remain blocked for the instrument to detect a note being played?

[$\ln(3) = 1.09$, $\ln(10) = 2.303$]

- A. $2.426 RC$
- B. $1.213 RC$
- C. $3.393 RC$
- D. $0.6065 RC$

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Q21. Ruben's Tube: Consider the tube at a fixed point in time when the antinodes are at their maximum pressure. Assume that the amplitude of the pressure wave is 26.5 kPa, and this wave reflects back to form standing waves in the tube. Find outflow velocity (m/s) at the hole at the wave's antinode.

[$\sqrt{555.5} = 23.5$, take density of propane as 1.8 kg/m^3 , $P_{atm} = 101 \text{ kPa}$.]

- A. 47
- B. 23.5
- C. 94
- D. 33.2

Q22. Ruben's Tube: Say the tube has length 80 cm, and velocity of sound in propane is 240 m/s. Find the frequency of the harmonic for which there will be 5 nodes in the standing wave (including boundaries).

- A. 300 Hz
- B. 600 Hz
- C. 750 Hz
- D. 450 Hz

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Q23. Hydraphone: A performer presses a hydraphone key so that the displacement is given by $x(t) = X_0 \sin(\omega t)$. Find how the average intensity of sound produced varies with ω (angular frequency) of the sound wave produced, where the average intensity is the mean value of intensity over one full cycle.

- A. It depends only on the amplitude X_0 , and not on the frequency ω .
- B. It increases as ω increases.
- C. It decreases as ω increases, because the displacement term dominates at lower frequencies.
- D. It is independent of displacement, velocity, and acceleration, and depends only on the constants k_1, k_2, k_3

Q24. Singing Bowl: An alloy of 80% Cu and 20% Sn by mass is prepared to be used for coating the singing bowl. Assume 1.00 mol of alloy atoms is taken. If all the Sn atoms present are incorporated into the γ -Brass type phase of the alloy, approximately how many moles of γ -Brass type phase formula units can form? (Atomic masses: Cu = 63.5, Sn = 118.7)

- A. 0.023 mol
- B. 0.011 mol
- C. 0.056 mol
- D. 0.080 mol

Q25. AudioCube: Two AudioCubes are fixed at points A (0,4), B (6,0). A third cube C kept at point (x,y) moves on the circle $x^2+y^2=25$. The AudioCubes are programmed in such a manner that sound is only

Space for Rough Work

registered by the software if the distance of AC is equal to the distance of BC. How many places can the cube C be placed for sound to be registered?

- A. 0
 - B. 1
 - C. 2
 - D. Infinitely many
-

ANSWER KEY

1. B	2. B, D	3. A	4. C	5. B
6. C	7. A	8. D	9. B	10. D
11. D	12. A, C, D	13. A, C, D	14. D	15. D
16. D	17. A	18. A	19. A	20. B
21. A	22. B	23. B	24. B	25. C

SOLUTIONS**The Sound of Music****Q1.** (Answer: B)*Solution:* The octave starts from C, so the order of notes is:

....A4, B4, C5, D5, E5, F5, G5, A5....

So, the D# note of the A5 octave is D#5 which is 6 semitones above A4 so $440 * 1.414 = 622$ Hz

Q2. (Answer: B, D)

Solution: Only B, C, D are harmonics (integer multiples of 440), and only B, D are higher octaves (A5, A6)

Q3. (Answer: A. 7)

Solution:

Perfect fifth has frequency ratio of 3:2

$$2^{\left(\frac{n}{12}\right)} = 1.5$$

Log base 2 both sides, solve for n.

Q4. (Answer: C)

Solution:

Frequencies in geometric progression means they are in constant ratios. Logarithmic scale would convert a GP to an arithmetic progression, which is what the graph shows. Hence, it must be a logarithmic scale

Q5. (Answer: B)

Solution: A semitone step multiplies frequency by $2^{\frac{1}{12}}$. The absolute Hz difference is $(2^{\frac{1}{12}} - 1)f$ which grows with frequency. The *frequency ratio* is what stays constant as frequency changes.

Q6. (Answer: C. 415 Hz)

Solution:

One step is 4 semitones, so frequency ratio r is $2^{(4/12)}$.

So, ratio of top note to first note is $2^{(8/12)}$

$$261.63 \times 2^{\frac{8}{12}} = 415 \text{ Hz}$$

I Can't H(Ear) You

Q7. Answer: A

Solution: Sheets can provide high tensile strength due to multiple parallel hydrogen bonds.

Q8. Answer: D

Solution: Only (D) has a lone pair other than the nitrogen and oxygens of the amino and carboxylic acid groups.

Q9. Answer: B

Solution:

A clearly wrong because proteins can't just break into amino acids from vibration

C clearly wrong because potassium can't just run out as ion pumps continuously supply more

D will be tricky for students because it is true by itself, but the question is clearly asking about damage to nerve fibers not the external ear (eardrum).

Q10. Answer: D - very clearly explained in paragraph that glutamate helps in transmission of signal

Q11. Answer: D

Solution:

$$f_2/f_1 = 2^{(6/12)} = 2^{(1/2)}$$

$$e^{6.909(x_1 - x_2)} = 2^{(1/2)}$$

$$\Delta x = \ln(2) / (2 * 6.909)$$

Q12. Answer: A, C, D

Solution:

A: True

Curve A intersects the X-axis since $I=0$ when membrane potential = equilibrium potential.

B: False

After the ATP pump is inhibited and gradient collapses, the inside and outside $[K^+]$ becomes exactly equal due to

passive ion transport. Therefore, equilibrium potential shifts towards 0 mV.

C: True

True, if membrane potential < equilibrium potential, the driving force is negative and hence K^+ ions rush inwards.

D: True

The slope of I-V graph (conductance) does not depend on the concentration difference of ions, therefore it remains constant. Only equilibrium potential shifts on the inhibition of ATP pump.

I CAN HEAR YOU NOW!

Q13. (Answer: A, C, D)

Solution: A is obviously true (without any charge on capacitor, there won't be any voltage across it even as capacitance varies).

Dynamic microphones are *less sensitive* because they have a heavier diaphragm which doesn't move as easily as that of a condenser mic.

C, D are both correct, explanation is in the option itself.

Q14. (Answer: D)

Solution:

1. Power gain is square of voltage gain so power gain is 16.
2. Gain in decibels is $10 \log(16) = 40 \log(2) = 12$ (approx.)
3. If output power was doubled, power ratio would be 32 instead of 16. Then gain in decibels would be $10 \log(32) = 10 \log(16) + 10 \log(2)$ therefore the increase in gain would be 3 dB.

Q15. (Answer: D)

Solution:

Both fundamentals are below 400 Hz so they are unchanged. The harmonic of 220 Hz (first string) is at 440 Hz (i.e. between 400-600 Hz), so it is reduced by 10 dB. The harmonic of 330 Hz (second string) is at 660 Hz, so it is reduced by 12 dB

Q16. (Answer: D)

Solution: Each coefficient should increase by a multiplicative factor of α .

Total delay between original signal and 2nd copy is τ so delay of first copy should be $\tau / 2$.

Q17. (Answer: A. 0.3162)

Solution:

For lowest value of α , there would be -20 dB attenuation (22 dB to 2 dB) in the $\alpha^2 x(t - \tau)$ copy of signal compared to original signal.

$$-20 \text{ dB} = \log(P_2/P_1) = 20 \log(\alpha^2) = 40 \log(\alpha)$$

$$\log(\alpha) = -0.5 \text{ so } \alpha = \frac{\sqrt{10}}{10} = 0.3162$$

Note: α^2 is the amplitude ratio, power ratio is *square* of that. Taking power ratio as α^2 leads to $\alpha = 0.1$ which is wrong.

Q18. (Answer: A)

Solution: Natural frequency: $\omega = \sqrt{\frac{k}{m}}$. So, A is the subwoofer and B is the tweeter.

Tweeter plays higher frequencies so requires a high pass filter before it. Similarly, a subwoofer requires a low pass filter before it as it plays lower frequencies.

YOU WOULDN'T HAVE HEARD THESE

Q19.

(Answer: A. 4mm)

Solution:

Firstly, we need to see whether f_2 is 280 KHz or 240 KHz (absolute value of difference is 20 KHz in both cases). For this, use the hint in the question. Moving hand further from antenna $\rightarrow d$ increases $\rightarrow C$ decreases $\rightarrow f_2$ increases. If frequency of sound *decreases* as f_2 increases it must mean that $f_1 > f_2$ so that $f_1 - f_2$ decreases as f_2 increases.

So now we have f_2 is 240 KHz. Using the standard formulae we can find:

$$C = \frac{1}{(2\pi \times 240,000)^2 \times 0.01} = \frac{1}{(150,720)^2} \approx 5\epsilon_0 \text{ (from frequency of LC oscillation formula)}$$

$$\text{So } d = \epsilon_0 A/C = A/5 = 4 \text{ mm.}$$

Q20.

(Answer: B)

Solution:

Using KVL:

$$-iR - Q/C = 0 \text{ or } dQ/dt = -Q/RC$$

Q on capacitor becomes:

$$Q(t) = Q_0 e^{-t/\tau} \text{ where } \tau = RC.$$

Thus voltage output: $V(t) = V_0 e^{-t/\tau}$. For comparator to be triggered, $V(t) = 0.3 V_0$

$$\text{So, } -t/\tau = \ln(0.3) \text{ or } t = \tau \times [\ln(10) - \ln(3)] = 1.213 RC$$

Q21.

(Answer: A. 47)

Solution:

Wave amplitude is 26.5 kPa, when it reflects and forms a standing wave, then amplitude of *that* will be double of original amplitude, hence 53 kPa.

At the antinode, pressure inside will be $50 + 53 \text{ kPa} = 103 \text{ kPa}$. Thus the pressure difference from inside to outside is 2 kPa. Thus, from Bernoulli's formula:

$$v = \sqrt{\frac{2(P_{in} - P_{atm})}{\rho}} = \sqrt{\frac{4000}{1.8}} = \sqrt{2222} = \sqrt{555.5 \times 4} = 23.5 \times 2 = 47 \text{ m/s}$$

Q22.

(Answer: B. 600 Hz)

Solution:

5 nodes $\rightarrow$ 4 antinodes $\rightarrow$ 4th harmonic.

Frequency of 4th harmonic is $4 \times \frac{v}{2L} = 4 \times \frac{240}{1.6} = 600 \text{ Hz}$

Q23.

(Answer B)

Solution: $I_{avg} = 1/T \times \int_0^T I(t) dt$, and then differentiate I_{avg} with respect to ω to get that term directly proportional to ω , $\omega > 0$ therefore I_{avg} increases with ω

Q24.

(Answer B)

Solution:

Calculate the moles of Sn atoms

Total moles of Cu + Sn = 1.00

Mass fraction of Sn = 0.20

$$(n_{\text{Sn}} \cdot M_{\text{Sn}}) / ((n_{\text{Cu}} \cdot M_{\text{Cu}}) + (n_{\text{Sn}} \cdot M_{\text{Sn}})) = 0.20$$

Put $n_{\text{Cu}} = 1 - n_{\text{Sn}}$, $M_{\text{Sn}} = 118.7$, $M_{\text{Cu}} = 63.5$ in the above equation to get $n_{\text{Sn}} = 0.1180$ (approx).

1 mol of $\text{Cu}_{41}\text{Sn}_{11}$ (γ -Brass type phase formula unit) requires 11 moles of Sn. Therefore, 0.1180 moles of Sn will form $0.1180/11$ moles of $\text{Cu}_{41}\text{Sn}_{11}$

$$0.1180/11 = 0.01072 \text{ mol} = 0.011 \text{ mol (approx)}$$

Q25.

(Answer C)

Solution:

Find the equation of locus of the perpendicular bisector of AB.

Slope of AB = $-2/3$.

Slope of perpendicular bisector = $3/2$

Midpoint of AB = (3,2)

Using slope-point form, the equation of the perpendicular bisector is: $3x - 2y - 5 = 0$

Since the third cube is kept on the circle, the equation of the perpendicular bisector must satisfy the equation of the circle. Put $y = (3x-5)/2$ into the equation of the circle and find the discriminant of the quadratic equation in x.

Since Discriminant > 0 , there are two real and distinct solutions of x and hence two such points for C to be placed.